How a Cargo Ship Works

by ReadWorks



It's 4:55 on a sunny afternoon 62 miles off the coast of Los Angeles, and the cargo ship *Mahimahi* is sailing southwest on its regular trip to Honolulu. The *Mahimahi* is big. It's 860 feet long, and right now it's carrying more than 30,000 tons, which is close to the maximum it can carry. Fully loaded, its hull is filled with shipping containers that rise 45 feet above the main deck.

When a ship this large is traveling like this, cruising at 18 knots an hour in a calm and pleasant sea (that's about 20 miles an hour), it looks as though nothing could stop it. But actually, powerful forces are at work, trying to blow the *Mahimahi* off course.

To understand how, first you need to know a few things about ships. When the captain stands at the wheel and looks to the front of the ship, the starboard is to his right. The port side is to his left. (Here's an easy trick to remember this: both "port" and "left" have four letters.) The front of the ship is called the bow, and the back is called the stern.

So right now the waves are sweeping in from the north, which means they are hitting the *Mahimahi* on the starboard side. The waves are medium sized, just under ten feet. Meanwhile the wind comes from the west at ten knots, hitting the ship's starboard bow.

Since the *Mahimahi* is so big, you might think such weak current and wind make no difference. Actually, they can change everything. How they do it involves the ship's pivot point. Let's say that you accidentally trip your friend as he's walking down the sidewalk. If you bump into his chest, that would just shove him sideways. Instead, your friend trips over your foot.

Why does your friend trip? Because when your friend is walking forward he carries most of his mass in his chest and his belly, so bumping into him there doesn't do much. But his feet don't

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weigh very much and they're the farthest thing away from his chest, so tripping him may lead to a big fall.

Well right now, the wind and current are trying their best to trip the *Mahimahi* by its ankle. As the ship moves forward, the center of its mass moves toward the bow. This is called its pivot point, because the whole ship wants to spin-or pivot-around it. The wind and waves hit this spot, and if they're strong enough they might succeed in shoving the ship sideways. But they're not going to catch the *Mahimahi* off guard and spin it around.

Where the ship feels the most pressure is at its stern. The very back of the ship is like its feet -the farthest spot away from its center of mass. As the *Mahimahi* sails southwest, therefore, the current is trying to grab its tail and spin it. If the current wins, the *Mahimahi* will spin out of control.

Meanwhile, the wind is trying to help the current spin the ship. The wind has little effect up front on the pivot point, but it's also blowing hard on the starboard stern. If the *Mahimahi* lost power, the wind pushing on the stern would start the ship in a clockwise spin. And then the current would pile on and continue the spin until the whole ship is pointing north. Even now, when the *Mahimahi* has full power, the wind sweeping against the bow is slowing the ship down.

All of this presents a problem, especially for Matson, the shipping company that owns the *Mahimahi*. Today is Saturday. Matson has promised its customers the *Mahimahi* will arrive in Honolulu next Wednesday, at midnight. To make a trip of 2,560 miles in four and a half days, the *Mahimahi* must keep its nose on course and its speed steady.

How is the ship's captain keeping a straight course? By using his controls to fight the elements. To keep the ship going straight, he keeps the rudder turned a little to starboard, pushing the bow slightly into the wind. To maintain a speed of 18 knots, he pushes the engine a little harder, maybe to 18-and-a-half knots.

At 6:50 pm, 75 miles into its trip, the *Mahimahi* passes the rocky southern tip of San Nicolas Island. An ocean observation station on the island shows currents swirling clockwise, right into the ship's bow. As the *Mahimahi* hits the oncoming current, it slows to 17.7 knots. A few minutes later the ship sails forward into calmer water, and it returns to its original speed. A person standing on the island's shore would never notice the difference.

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